

MODEL METRICS: CALIFORNIA ELECTRICITY DEMAND FORECASTING

MODEL IDENTIFICATION

- Final Model Type: Linear Regression.
- Reason for Selection: Linear Regression achieved the strongest validation performance across MAE, RMSE, and R^2 while remaining highly interpretable, lightweight, and easy to deploy in the Streamlit forecasting application. Although more flexible ensemble methods were evaluated, the simpler linear model generalized better on unseen data and captured the core weather-demand relationships effectively.

HYPERPARAMETERS

- LinearRegression(): scikit-learn default parameters.
- Numeric preprocessing: StandardScaler().
- Categorical preprocessing: OneHotEncoder(handle_unknown="ignore").
- Pipeline framework: Scikit-learn Pipeline and ColumnTransformer.
- Chronological train/validation/test split:
- Train: 70%.
- Validation: 15%.
- Test: 15%.

PERFORMANCE METRICS

- Validation MAE: 564.93 MW
- Validation RMSE: 731.58 MW
- Validation R^2 : 0.9233
- Test MAE: 596.91 MW
- Test RMSE: 765.35 MW
- Test R^2 : 0.8874

MODEL COMPARISON SUMMARY

- Linear Regression:
 - Validation MAE: 564.93
 - Validation RMSE: 731.58
 - Validation R^2 : 0.9233
- Gradient Boosting Regressor:
 - Validation MAE: 640.43
 - Validation RMSE: 806.99
 - Validation R^2 : 0.9067
- Ridge Regression
 - Validation MAE: 645.86
 - Validation RMSE: 844.79
 - Validation R^2 : 0.8977
- Random Forest Regressor:
 - Validation MAE: 778.70
 - Validation RMSE: 941.07

- Validation R^2 : 0.8731

FINAL MODEL FEATURES

- The final model utilized 29 engineered numeric features derived from statewide weather summaries, lagged demand behavior, calendar structure, and interaction terms.
- Lag / Persistence Feature:
 - lag1_load: Previous day California load.
- Aggregated Weather Features:
 - temp_mean_all: Mean statewide temperature.
 - temp_max_all: Mean statewide maximum temperature.
 - temp_min_all: Mean statewide minimum temperature.
 - temp_range_all: Daily statewide temperature range.
 - rhum_mean_all: Mean statewide relative humidity.
 - prcp_sum_all: Statewide precipitation total.
 - wspd_mean_all: Mean statewide wind speed.
- Nonlinear Temperature Features:
 - cooling_degree
 - heating_degree
 - cooling_degree_sq
 - heating_degree_sq
 - is_hot
 - is_very_hot
- Calendar Features:
 - is_weekend
 - is_summer
 - is_winter
 - is_shoulders_season
- Cyclical Calendar Encodings:
 - month_sin.
 - month_cos.
 - day_of_week_sin.
 - day_of_week_cos.
 - day_of_year_sin.
 - day_of_year_cos.
- Interaction Features:
 - hot_weekend_interaction.
 - cold_weekend_interaction.
 - summer_cooling_interaction.
 - winter_heating_interaction.
 - humidity_heat_interaction.

MODEL INSIGHTS

- Temperature was the dominant driver of electricity demand, especially during warmer periods associated with increased cooling usage.

- Lagged demand significantly improved forecasting performance, indicating strong short-term persistence in California electricity usage patterns.
- Aggregated statewide weather summaries produced more stable predictive signals than relying on any single city independently.
- Calendar-aware feature engineering substantially improved model performance by capturing weekly and seasonal structure.
- Despite testing more flexible ensemble models, Linear Regression generalized best due to the strong engineered feature set and relatively structured relationships in the data.

DEPLOYMENT SUMMARY

- The final trained Linear Regression pipeline was deployed in a Streamlit forecasting application.
- The app retrieves:
 - Live weather forecasts from OpenWeatherMap.
 - Recent CAISO load data through GridStatus.
- The forecasting system reconstructs the same preprocessing and feature engineering pipeline used during training before generating recursive 4-day electricity demand forecasts.
- Maintaining consistency between training-time and inference-time feature engineering was critical for producing realistic forecast outputs.